

Traian Florin Șerbănuță

Formal Methods for Software Correctness · Verification Engineer · Associate Professor

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Education

PhD in Computer Science <i>University of Illinois, Urbana-Champaign</i>	2010
Master in Computer Science <i>University of Bucharest</i>	2004
Bachelor in Computer Science <i>University of Bucharest</i>	2002

Experience

Associate Professor of Computer Science <i>University of Bucharest, Faculty of Mathematics and Informatics</i> Designed and taught courses across software modelling, declarative & concurrent programming, programming-language semantics, program verification, and machine learning — all course materials openly published. Supervise graduate students and serve on departmental committees.	2013–Present
Consultant and Researcher <i>Asymptotic</i> Prototyping AI-assisted, Lean-based provers for programming languages. Lead author of <i>Rust-Prover</i> , a Lean 4-backed verifier for Rust: specifications and Rust code are translated into Lean theorems that LLM agents prove, with Lean’s kernel as the final check. It proved all 1325 theorems of the 1007-problem VeriContest benchmark.	2026–Present
Consultant and Researcher <i>Pi Squared, Inc.</i> Core Rust engineer on Pi Squared’s verifiable-computing and universal-settlement infrastructure (“Proof of Proof”). Work spans five phases of the platform’s evolution: • Pi² research prototype (2024) — Metamath proof checkers compiled to five zkVM backends (RISC Zero, SP1, Nexus, Lurk, Delphinus), with driver tooling and cross-backend benchmarking. • Blocks to Circom ZK pipeline (2025) — Rust toolchain compiling the Blocks DSL into Circom subcircuits and generating end-to-end ZK certificates for block instantiations. • Verifiable Settlement Layer (2025) — claim-based settlement layer where submitted claims reach quorum and are recorded as signed settlements. • FastSet validator and proxy (2025–2026) — validator and client-facing proxy for the FastSet protocol. • Fast Shop / agentic AI commerce (2026) — Universal Commerce Protocol integration with region-aware multi-region delivery, a Shopify backend, and an MCP-exposed commerce surface.	2024–2026

Consultant and Researcher

2012–2023

Runtime Verification, Inc.

Long-running principal contributor across four projects:

- **K Framework — Java implementation (2012–2015)** — executor/debugger internals for the Maude and Java backends: LTL model-checking, search-graph extraction, and AC matching.
- **RV-Predict — race & deadlock detector (2012–2017)** — maximal-causal-model predictive race detection for Java and C/C++ (via an LLVM instrumentation pass), scaled to a 1M-variable cap.
- **K Haskell Backend — symbolic-execution prover (2017–2022)** — all-path/one-path reachability-logic prover with SMT integration, applied to the formal semantics of the EVM, WebAssembly, and IELE.
- **CBC-Casper / VLSM consensus in Coq (2022–2023)** — Coq modelling and verification of blockchain protocols, including one of the first safety formalizations of CBC Casper.

Postdoctoral Research Fellow

2011–2013

Alexandru Ioan Cuza University, Iași (FMSE Laboratory)

Postdoctoral Research Associate

2011–2012

University of Illinois, Urbana-Champaign (Information Trust Institute)

Research Assistant

2004–2010

University of Illinois, Urbana-Champaign (FSL Laboratory)

Teaching Assistant

2003–2004

University of Bucharest, Department of Computer Science Fundamentals

Summer Intern

2007

Google, New York

Summer Intern

2005

Microsoft Research, Redmond (Testing, Verification, and Measurement Group)

Programmer

2000–2001

Popnet-Agentscape Romania, Natural Language Processing Team

Open-source Projects

Personal research and tooling outside employer-affiliated work; full list at <https://github.com/traiansf>.

- **Formalization libraries** — aml-in-coq (Applicative Matching Logic), arl-in-coq (Abstract Rewrite Systems), sets-in-coq.
- **Type theory study** — propositions-as-types, Coq scribbles along *Type Theory and Formal Proofs* (Nederpelt & Geuvers).
- **Teaching companions** — semantics-in-coq and semantics-in-lean, companions to the Foundations of Programming Languages course.
- **Teaching tools** — multiple_choice_exam, built with AI coding agents: a Python generator for randomized, print-ready exam variants from a Markdown question bank, and a Dart mobile app that grades answer sheets by QR code and optical mark recognition.
- **Schools & events** — bucharest-lean-ac, Bucharest Autumn School materials in Lean 4.
- **Other** — excel-database, a WordPress plugin (★4) that exposes an Excel spreadsheet table as queryable data.

Skills

Formal methods and verification Programming language semantics AI-assisted software development
LLM agents for theorem proving Rewriting logic K Framework Temporal logics Model checking
Specification and verification techniques

Programming Languages

Rust Haskell Coq / Rocq Java C / C++ K / Maude Python Lean 4 Prolog

JavaScript / TypeScript

PHP

Languages

Romanian (native) · English (fluent) · French (basic)

Publications

55 articles · h-index **22** · **2178** citations (Google Scholar, as of October 2026)